Social and behavioral predictors of diabetes

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Summary of research questions

1. *Does household income affect the prevalence of diabetes?*

   The study used demographic and health outcomes data to investigate whether low-income groups have a higher chance of having diabetes, which may reflect the impact of differences in health care and lifestyle on diabetes prevalence.

2. *Can we build a machine learning model to predict diabetes based on labels such as social behavior?*

   The question was to investigate whether data such as BMI, smoking status, and sleep duration could be used to predict diabetes risk, and how accurately the prediction could be made in the absence of medical testing.

3. *Which factors are most important in predicting diabetes risk?*

   This question analyzes the importance of predictive factors such as BMI, sleep duration, and smoking status on diabetes status, providing insights into diabetes prediction and prevention strategies.

4. *Do individuals from different backgrounds have different rates of participation in health checkups?*

   This question explores differences in individual health care behaviors across different contexts and potential barriers faced by different individuals in accessing health services.

Motivation

My father has diabetes, which is one reason that motivates me to explore and predict the social and behavioral factors that may affect the risk of diabetes. In this project, I will combine real-world health and demographic data to explore how income, lifestyle, and different backgrounds affect the prevalence of diabetes. Understanding and predicting factors that increase the risk of diabetes can provide recommendations for policy and health education efforts, thereby improving public health.

Dataset

1. NHANES 2017–2020 Demographics

   URL: https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Demographics&CycleBeginYear=2017

   This dataset provides demographic information such as age, gender, income, education, and nationality. The data is collected by the National Center for Health Statistics of the Centers for Disease Control and Prevention (CDC).

2. Diabetes Health Indicators Dataset

   URL: https://www.kaggle.com/datasets/alexteboul/diabetes-health-indicators-dataset

   The dataset provides data on lifestyle and health indicators (BMI, smoking status, sleep duration, etc.) as well as diabetes outcomes for more than 250,000 individuals.

I will join these datasets using relevant demographic columns such as age and gender.

Challenge goals

1. ***Multiple Datasets***

   This project will combine the NHANES demographic data set and the BRFSS diabetes health indicator data set for multidimensional analysis. I will use join operations to combine demographic and lifestyle characteristics for analysis and explore the relationship between the two data sets for at least two research questions.

2. ***Machine Learning***

   For this project, I will build and evaluate a machine learning model to predict diabetes outcomes based on behavioral and demographic data. I will compare model performance under different hyperparameters and evaluate feature importance to identify the predictors that are likely to be most influential.

Method

1. *Method for Research Question 1*
   - Clean and filter both datasets to include individuals with complete income and diabetes information.

- Use .groupby('income_level')['diabetes'].mean() to calculate diabetes rates by different income levels.
- Visualize the relationship using Seaborn.
- Standardize income levels between datasets before merging.

2. *Method for Research Question 2*
    - Use BRFSS dataset features, such as BMI, physical activity, and mental health, etc. to train the model.
    - Preprocess the data to handle missing values and normalize features.
    - Use sklearn models.
    - Evaluate the accuracy of the model on the test set.

3. *Method for Research Question 3*
    - Use model feature importance scores.
    - Visualize using a bar chart sorted by importance.
    - Analyze which variables consistently rank high in the model.

4. *Method for Research Question 4*
    - Use NHANES nationality and health service usage columns to analyze.
    - Compare checkup rates across groups using .groupby('nationality')['usage'].mean().
    - Discuss barriers that individuals from different backgrounds may face.

EDA Results

1. Dataset Size
    - NHANES: 9254 rows x 46 columns. Each row represents an individual with various behavioral and health indicators.
    - BRFSS: 253,680 rows x 22 columns. Each row represents an individual with demographic attributes.
    -
```
Nhanes shape:  (9254, 46)
BRFSS shape: (253680, 22)
```
    - Joined Dataset: After merging ID, age, and gender, the dataset contains 89 rows, and after removing missing values, the number is reduced to 87. This suggests that ID is not a shared identifier between datasets, a problem I plan to address in future work by relaxing join conditions and other strategies. In future analyses, I plan to modify the join strategy, such as using only age and gender, which would significantly increase the sample size.
    -
```
Merged shape: (89, 7)
Cleaned shape: (87, 7)
```

2. Missing Value

- NHANES: The household income column has some missing values. The missing percentage is less than 5%. I used dropna() to remove rows containing missing values for the key variable.
- BRFSS: The main analysis columns, such as Diabetes_012, BMI, and Income, are not missing.
- Joined: Use .dropna() to remove missing records of key variables such as Age, Sex, Income, Diabetes status, and BMI to ensure the reliability of the analysis.

```
merged = merged.dropna()
print('Cleaned shape:', merged.shape)
```

-
```
Merged shape: (89, 7)
Cleaned shape: (87, 7)
```

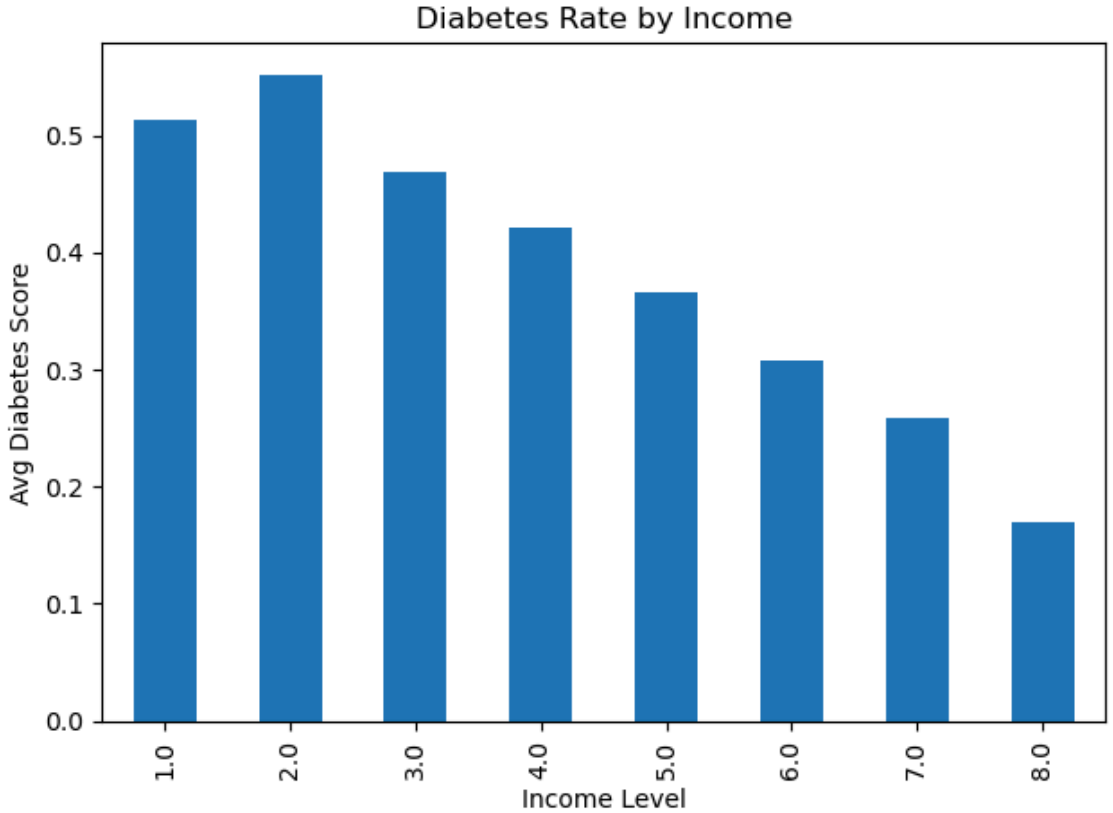
3. Variable

Variable	Meaning	Source
Diabates_012	0 = No diabetes, 1 = Pre-diabetes, 2 = Diabetes	BRFSS
Income_x	Household income level	BRFSS
BMI	Body Mass Index	BRFSS
Age	Age	BRFSS & NHANES
Sex	Gender	BRFSS & NHANES

4. Seven Number Summary

```
BMI Summary:
count    2.249856e+07
mean     2.838711e+01
std      6.624942e+00
min      1.200000e+01
25%      2.400000e+01
50%      2.700000e+01
75%      3.100000e+01
max      9.800000e+01
```

5. Data Visualization



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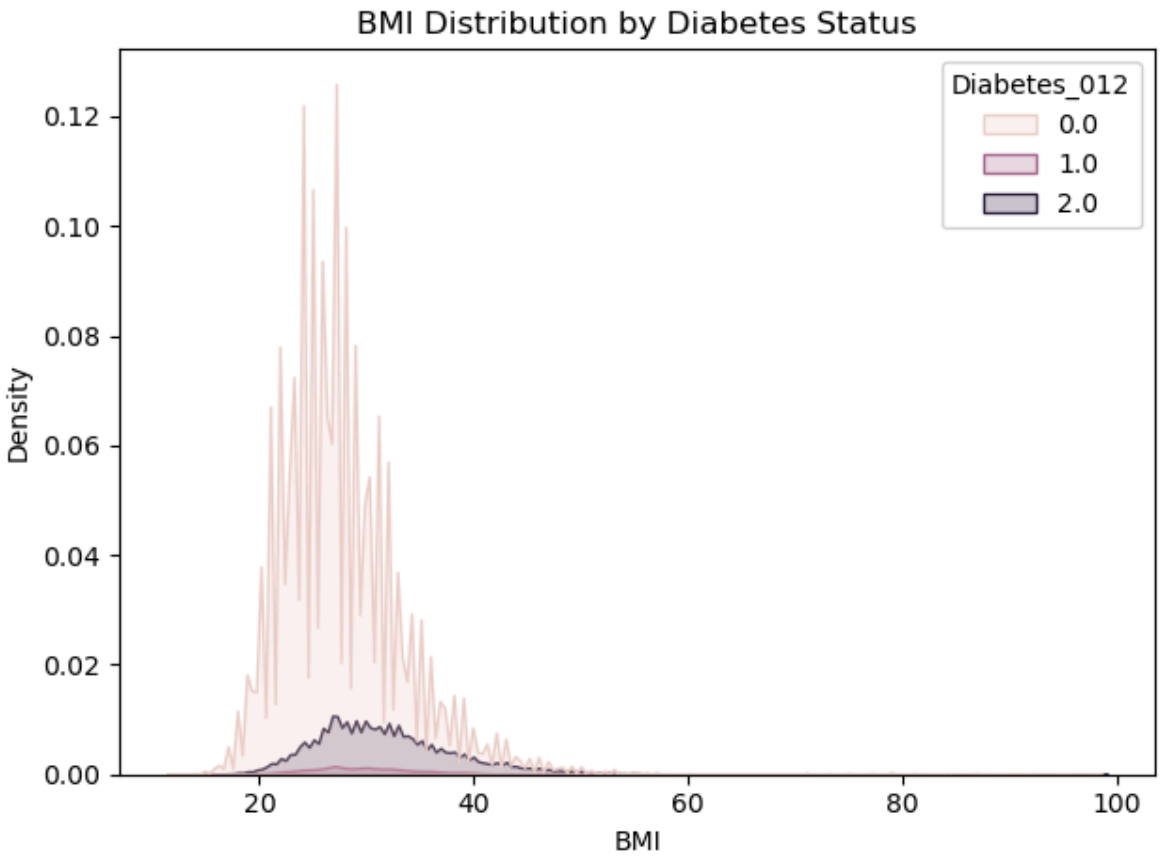
- This figure shows the average diabetes scores across different income groups. A clear downward trend can be seen. Individuals in the lowest income group (Level 1) have the highest average diabetes scores, while those in the highest income group (Level 8) have the lowest. This supports the idea that low-income individuals may be at greater risk for diabetes, potentially due to differences in access to healthcare, diet, or lifestyle factors.



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- This figure shows the distribution of BMI by diabetes status. The non-diabetes group (0) has a lower overall BMI, peaking around 25. The diabetic group (2) has a significantly higher BMI distribution, peaking near 30-35. This visualization reinforces the association between obesity and diabetes risk and emphasizes that BMI is an important predictor in our analysis.

6. Relationship Between Datasets
    - This project uses the following datasets:
    - NHANES 2017–2020 Demographic Data Set (from the CDC): Contains demographic information such as age, sex, and income.
    - BRFSS 2015 Diabetes Health Indicators Dataset (from Kaggle): Contains behavioral and health-related data, including diabetes status, BMI, smoking habits, and physical activity.
    - While these datasets are not from the same individuals, they share some common columns, such as age and sex. This allows us to merge them based on their shared demographic attributes, creating a synthetic yet consistent dataset that links demographic characteristics to health behaviors.
    - The combined dataset allows us to explore the relationship between demographic factors, such as income (from NHANES), and diabetes outcomes (from BRFSS), which would not be possible using either dataset alone.

Work Plan

Task	Estimated Time	Actual Time
Dataset downloading and basic cleaning	2 hours	~30 minutes
Merge datasets and preprocess features	4 hours	~1 hours
Data Analysis for Research Questions 1 and 4	5 hours	~3 hours
Machine Learning Modeling for Research Questions 2 and 3	10 hours	Not Yet
Report	5 hours	Not Yet
Visualization and final presentation	6 hours	Not Yet

I plan to complete all the tasks by myself. I will do all the development in vscode. I will write modular code. I will test functions step by step and comment on the code. If I encounter unexpected problems when processing the data, I will simplify the modeling task or remove non-critical variables. If there are unexpected challenges I'm not able to solve I will post the problem on ed or come to the office hours.

Test

To ensure the correctness of our EDA code and analysis, we implemented the following testing strategy:
Using assert statements to confirm that the cleaned dataset has no missing values:
assert merged_cleaned.isnull().sum().sum() == 0
Printing the dataset shape and using .head() to check the merge logic
Using .describe() to manually check the distribution using .value_counts()
Drawing all visualizations and visually verifying that:
X- and Y-axis labeling is correct
Presentation of a title and legend
Trends are consistent with expectations (e.g., higher rates of diabetes in low-income groups)